

Chapter 2: Solving systems using matrices Worksheet

1. Are each of the following statements true or false?

- (a) $\left[\begin{array}{cc|c} 2 & 8 & 6 \\ 2 & 3 & 2 \end{array} \right]$ The systems of linear equations represented by the augmented matrices $\left[\begin{array}{cc|c} 2 & 8 & 6 \\ 2 & 3 & 2 \end{array} \right]$ and $\left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & 1 & \frac{4}{5} \end{array} \right]$ have the same set of solutions.
- (b) If A is a 3×2 matrix, it's possible for the rank of A to be 3.
- (c) A homogeneous system of 2 equations in 3 unknowns always has nontrivial solutions.
- (d) If A is a 3×3 matrix with rank 2, then the homogeneous system with coefficient matrix given by A will have only the unique trivial solution.

Solution.

- (a) True. The second matrix is a row-echelon form of the first.
- (b) False. The rank of A is the number of leading ones in a row-echelon matrix row-equivalent to A . If there were 3 leading ones, two of them would “stack” over each other, making it not a row-echelon matrix.
- (c) True. If we perform Gaussian elimination on the 2-by-4 augmented matrix, then unless the system is inconsistent, we are guaranteed at most 2 pivots, and therefore at least 1 column corresponding to a free variable. A homogeneous system cannot be inconsistent, because it always has the trivial solution.
- (d) False, this system will actually have infinitely many solutions, since there will be a free variable after row-reducing A .

2. Perform row operations on each matrix to reduce it to a row echelon form. Afterwards, find the reduced row echelon form for the matrices. What is the rank of each?

$$(a) A = \begin{bmatrix} 0 & 1 \\ 1 & -2 \\ 2 & 0 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 2 & 2 & 2 & -2 \\ 4 & -5 & -5 & 5 \\ 0 & -4 & -4 & 4 \end{bmatrix}$$

Solution.

Note that there are multiple row echelon form matrices for each matrix. In contrast, each matrix has a unique reduced row echelon form matrix.

(a) Swap rows 1 and 3 to obtain

$$B = A_{\mathbf{r}_1 \leftrightarrow \mathbf{r}_3} = \begin{bmatrix} 2 & 0 \\ 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Multiply row 1 by $\frac{1}{2}$ to obtain

$$C = B_{\frac{1}{2}\mathbf{r}_1 \rightarrow \mathbf{r}_1} = \begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Add -1 times row 1 to row 2 to obtain

$$D = C_{-\mathbf{r}_1 + \mathbf{r}_2 \rightarrow \mathbf{r}_2} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix}.$$

Multiply row 2 by $-\frac{1}{2}$ to obtain

$$E = D_{-\frac{1}{2}\mathbf{r}_2 \rightarrow \mathbf{r}_2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

Continuing, add -1 times row 2 to row 3 to obtain

$$F = E_{2\mathbf{r}_2 \rightarrow \mathbf{r}_3 \rightarrow \mathbf{r}_3} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

This is a row echelon matrix, and in fact is also in reduced row echelon form, so this is the reduced row echelon form matrix for A . The rank of A is 2.

(b) Adding -2 times row 1 to row 2 of A yields

$$B = A_{-2\mathbf{r}_1 + \mathbf{r}_2 \rightarrow \mathbf{r}_2} = \begin{bmatrix} 2 & 2 & 2 & -2 \\ 0 & -9 & -9 & 9 \\ 0 & -4 & -4 & 4 \end{bmatrix}.$$

Multiplying the second row of B by $-\frac{1}{9}$ yields

$$C = B_{-\frac{1}{9}\mathbf{r}_2 \rightarrow \mathbf{r}_2} = \begin{bmatrix} 2 & 2 & 2 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & -4 & -4 & 4 \end{bmatrix}.$$

3. Solve each system of equations by 1.) putting it into an augmented matrix and 2.) Using Gaussian Elimination to get the system into row-echelon form and 3.) Extracting the solutions. If a system has no solutions, explain why. If a system has multiple solutions, describe the family of solutions in terms of a free variable/parameter.

$$(a) \quad \begin{cases} 4x + y + 3z = 5 \\ 2x - 2z = 4 \\ 8x + y - z = 9 \end{cases}$$

$$(b) \quad \begin{cases} 4x + y + 3z = 5 \\ 2x - 2z = 4 \\ 8x + y - z = 13 \end{cases}$$

Solution.

For all of the above, we will compute a row-echelon form and then backsubstitute.

$$(a) \text{ Our augmented matrix is } \left[\begin{array}{ccc|c} 4 & 1 & 3 & 5 \\ 2 & 0 & -2 & 4 \\ 8 & 1 & -1 & 9 \end{array} \right]. \text{ We divide the top row by 4 to get}$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 2 & 0 & -2 & 4 \\ 8 & 1 & -1 & 9 \end{array} \right]. \text{ We subtract twice the first row from the second row to get}$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 0 & -\frac{1}{2} & -\frac{7}{2} & \frac{3}{2} \\ 8 & 1 & -1 & 9 \end{array} \right]. \text{ Multiplying the second row by } -2 \text{ yields } \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 0 & 1 & 7 & -3 \\ 8 & 1 & -1 & 9 \end{array} \right].$$

$$\text{Subtracting 8 times the top row from the bottom row gives } \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 0 & 1 & 7 & -3 \\ 0 & -1 & -7 & -1 \end{array} \right]. \text{ Adding}$$

$$\text{the second row to the third row gives } \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 0 & 1 & 7 & -3 \\ 0 & 0 & 0 & -4 \end{array} \right], \text{ and since } 0 \neq -4, \text{ this system}$$

is inconsistent.

$$(b) \text{ Our augmented matrix is } \left[\begin{array}{ccc|c} 4 & 1 & 3 & 5 \\ 2 & 0 & -2 & 4 \\ 8 & 1 & -1 & 13 \end{array} \right]. \text{ We carry out exactly the same procedure as}$$

in part 2. Since the bottom-right entry started as 13, and we subtract first 10 and then 3

$$\text{from it, at the end, we get the matrix } \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{3}{4} & \frac{5}{4} \\ 0 & 1 & 7 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So $z = t$ is a free variable, $y = -3 - 7t$, and

$$x = \frac{5}{4} - \frac{3z}{4} - \frac{y}{4} = \frac{5}{4} - \frac{3t}{4} - \frac{-3-7t}{4} = 2 + t$$

4. Find a nontrivial solution of the homogeneous system represented by the augmented matrices. If there is no nontrivial solution, explain why.

(a) $\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 5 & 2 & -1 & 0 \end{array} \right]$

(b) $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 2 & 5 & 0 & 0 \\ -3 & 3 & 1 & 0 \end{array} \right]$

Solution.

In each case, we perform Gaussian Elimination.

(a) Subtracting 5 times the top row from the bottom row gives $\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & -18 & -11 & 0 \end{array} \right]$.

Dividing the bottom row by -18 gives $\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & 1 & \frac{11}{18} & 0 \end{array} \right]$. Subtracting 4 times the bottom row from the top row gives $\left[\begin{array}{ccc|c} 1 & 0 & \frac{-8}{18} & 0 \\ 0 & 1 & \frac{11}{18} & 0 \end{array} \right]$.

The third variable is therefore a free variable, so we can choose its value to be, for instance, 1. $x - \frac{8}{18} * 1 = 0$, so $x = \frac{8}{18}$, and $y + \frac{11}{18} * 1 = 0$ so $y = -\frac{11}{18}$. So a nontrivial

solution is given by $\begin{bmatrix} \frac{8}{18} \\ -\frac{11}{18} \\ 1 \end{bmatrix}$.

(b) We perform row operations to reduce the matrix to a row-echelon form such as:

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Since the final row has not cancelled out to 0, we will not be left with any free variables, so there will be no nontrivial solutions.

5. Find the general parametrized solution to the homogeneous system represented by the

augmented matrix $\left[\begin{array}{cccc|c} 1 & 1 & -1 & 5 & 0 \\ 2 & -1 & -14 & 1 & 0 \\ -3 & 1 & 19 & -3 & 0 \end{array} \right]$

Solution.

The top row already has a leading 1. Adding -2 times the top row to the middle row and 3

times the top row to the bottom row gives $\left[\begin{array}{cccc|c} 1 & 1 & -1 & 5 & 0 \\ 0 & -3 & -12 & -9 & 0 \\ 0 & 4 & 16 & 12 & 0 \end{array} \right]$. Dividing the middle row

by -3 yields $\left[\begin{array}{cccc|c} 1 & 1 & -1 & 5 & 0 \\ 0 & 1 & 4 & 3 & 0 \\ 0 & 4 & 16 & 12 & 0 \end{array} \right]$. Adding -4 times the middle row to the third row gives

$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 5 & 0 \\ 0 & 1 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$. Adding -1 times the middle row to the top row gives

$\left[\begin{array}{cccc|c} 1 & 0 & -5 & 2 & 0 \\ 0 & 1 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$ The third and fourth variables are free, so we see that the general solution

is $x_1 = 5s - 2t$, $x_2 = -4s - 3t$, $x_3 = s$, $x_4 = t$, written as

$$\begin{bmatrix} 5s - 2t \\ -4s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 5 \\ -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

6. Given the matrix $M = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$, find two 3×3 (coefficient) matrices A_1 and A_2 , each

with the above reduced row echelon form M , but $\left[\begin{array}{ccc|c} & & & 1 \\ & A_1 & & -2 \\ & & & 3 \end{array} \right]$ has no solution, and

$\left[\begin{array}{ccc|c} & & & 1 \\ & A_2 & & -2 \\ & & & 3 \end{array} \right]$ has infinitely many solutions.

Solution.

Let $A_1 = M$, then $[A_1|\mathbf{b}]$ for $\mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ has no solution (the last equation is $0 = 3$ which cannot be satisfied). Construct A_2 as follows

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 3R_1} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -2 \\ 3 & 0 & -3 & 3 \end{bmatrix} := A_2.$$

Now let $A_2 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 3 & 0 & -3 \end{bmatrix}$. By applying $R_3 \rightarrow R_3 - 3R_1$ to A_2 , we get back to M , and we

can see that the system $[A_2|\mathbf{b}]$ for $\mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ has infinitely many solutions: Let

$$z = t, x = 1 + t, y = -2 - 2t.$$